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To Whom It May Concern:

I am writing in response to a “Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan”, issued by the National Science Foundation (NSF) on February 6, 2025. Until 2015, I was employed by the National Academy of Sciences for 30 years in Washington, DC with my last position there as founding director of the Board on Research Data and Information (2008-2015). I am now retired and a part-time data and information policy consultant to international organizations, governments, and universities. My complete professional profile is available on my website at: www.paulfuhler.com.

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Because my background and experience focused on the policy and management of research data and not the technical aspects of AI (for the sake of simplicity, I include Generative Artificial Intelligence and Artificial General Intelligence in that term), I will focus my remarks on the policy issues and not the technical ones. I also concentrate on the processes that need to be involved rather than on the substance of the issues, primarily because this is a very fast-moving and shifting area. The establishment of a flexible and quickly responsive process is essential to successfully implementing any AI Action Plan. And lastly, my remarks focus on the threats or concerns of the rapidly emerging technology, since the positive aspects and opportunities are better known and require less consideration.

With regard to the process, it is clear that the Trump Administration does not wish to burden industry with onerous regulations or requirements or to hamper the progress in developing AI. I share that concern, but we live in an interdependent global environment—whether we like it or not—so we need to cooperate with the international community for our own security and the world’s. It is not an overstatement to say that we are fundamentally at the dawn of a new era—another true paradigm shift—building on, but completely transformative of, the digital networks and the WorldWideWeb that began 30 years ago, and which constituted a paradigm shift in their own right.

I am not a technical expert, but I do know that AI requires massive amounts of factually accurate data and information to succeed. The data-intensive sciences to which we have become accustomed will be the fuel that drives AI and its limitless applications. What is especially important is that all fields of science, whether in the life sciences, physical sciences,

social sciences, engineering, the humanities—indeed, all areas of human endeavor—are poised to welcome the benefits.

Many countries already have huge, factually accurate, and georeferenced databases that will be exploited by AI. We can use them to make incredible advances, but if the advances we are about to make are unlimited and profound, so are the dangers posed by the technology that world leaders—especially in the development of AI technology itself—are warning us about. In many ways, the technical advances and their unlimited applications will be the easy part, although their successful implementation will require a great deal of effort and expense.

What concerns me as an information lawyer are the human systems that are used to control the technology and to avoid the worst pitfalls. The pace of technological progress will always outrun the capability of human systems to adequately control them. Moreover, the rapidity of technological advances is accelerating, making our ability to control them increasingly difficult. With AI we very well may be entering into an era of runaway technology, in which we are predicted to end up being the spectators and ultimately perhaps the victims of our own success. It is the vision called Singularity, which may only be two decades from now.

It therefore is not too early—and perhaps in many ways it is too late—to think seriously and to devote adequate resources for that control. We need to devise human systems to safely harness that technology. For that, AI systems themselves may be the answer, at least in part. Indeed, if asking the right question has been a hallmark of scientific advances in the past, that is now one of the most important skills for an AI-enabled approach to problem-solving and perhaps control.

As someone involved in science and innovation throughout my career, and in the data activities that have been both the spark and the fire, my first instinct would be to break the perceived problems into manageable pieces and to address the most important parts first. It is time-consuming and difficult enough to get national, much less global agreement on even the most mundane and simplest concerns. However, we must now facilitate a process that involves many different types of people, including political and business leaders, cyber-engineering experts, domain scientists, and the ethicists and lawyers from all countries, to work together in a sustained effort. We need to mute the worst potential effects of AI and to make progress on well-defined baskets of issues that are of near-term and longer-term focus.

Such attention should be concentrated, simultaneously, at the national and international levels. Separate, yet interrelated agreements though formal treaties and protocols should be devised, if possible, although some would say that it is a fool's errand if that is to be accomplished rapidly. The dismal state of international law in confronting data and other technological challenges is not inspiring.

We should endeavor to establish processes—now—that address a well-curated and continuously updated list of the most important problems. We need to devise methods to maintain control over the growing super-intelligences we are unleashing. The same or similar

processes should be set up in major domains of technical activity, such as biotechnology, nanotechnology, robotics, and so on. Together, these domains of our existence need to be coordinated in a manner that puts a premium on the safest results.

Some of the likely potential problems in the not-too-distant future are familiar while others may be less so. In fact, there are a number of well-known examples that have already occurred, even without the assistance of AI technologies. Well-publicized ones affect all areas of human endeavor, such as:

First, the massive dislocation of workforces in a broad range of sectors. This is not just manual labor in the service and manufacturing industries, but includes professional service sectors such as accounting, law, and medicine;

Second, the potentially damaging impacts on privacy protections by finding and processing personal data, or by using sensitive information for social surveillance. The EU has already enacted the GDPR on a regional basis and that may be a treaty that could be expanded more globally;

Third, greatly enabled criminal activities and capacities, especially cybercrimes, through various software attacks, data theft, malicious code development, or fabricated information. Many examples of that have already occurred, not just in the United States, but all over the world;

Fourth, an expansion of human-induced disasters and infrastructure attacks, such as shutting down electrical grids, incapacitating computer systems and networks, or breaching large-scale energy sources;

Fifth, the automation and enhancement of weaponry in warfare, and their hijacking by non-state actors in terrorism;

Sixth, the growth of inequality and the concentration of power. This is already happening in the US and throughout the world;

Seventh, and related to my previous point, the adverse effects from a growing gap between the highly developed economies and the poorer nations in AI capacities on a global basis. This too has been taking place for several decades, known as the digital divide, and

Eighth, extreme financial manipulation of markets and the propagation of economic instabilities.

The list goes on.

In the somewhat longer term, however, we are likely to see not only an expansion of many of these impacts, but also to experience entirely new threats that humanity must realistically be able to confront. Such global problems could be generated by the technological advances

themselves, or they could be geopolitical in nature, that is, a byproduct of the advancing technology.

In the arena of the technology itself, we could experience losing control to a self-aware super-intelligence that some well-informed experts have said might even lead to human extinction. Recent public statements of cyber-technology leaders, again in the US, have been alarming (Sam Altman, Geoffrey Hinton, Elon Musk, and many more).

How do we deal with such an existential crisis that may be confronting us? Humanity must develop and implement fail-safe mechanisms for that potential threat. The development of this AI Action Plan should have a role to play in that.

On the geopolitical front, we could easily exceed the boundaries of individual nations, leading to large-scale conflicts that may be very difficult to manage. What will be the processes that can be instituted, that can confront such eventualities and build trust?

In either case, experts in government, industry, and academia should become involved and work for our collective future. More specifically, we need to take some actions, such as:

- be proactive for safety and propose adequate mechanisms and potential solutions in the AI technology sector, sooner rather than later;
- increase research in safety mechanisms and improve prediction capabilities;
- work on standards in the appropriate organizations that promote common and effective approaches to technical and management issues;
- train employees in all sectors to use AI responsibly and promote transparency in its development and implementation;
- collaborate on ethical guidelines and insist on incorporating them in institutional and national policies, and international instruments, and last but not least;
- help ensure that humans always maintain supremacy over AI technology.

Best wishes,

Paul F. Uhler